

Chapter 13

Aviation Weather Services

Introduction

In aviation, weather service is a combined effort of the National Weather Service (NWS), Federal Aviation Administration (FAA), Department of Defense (DOD), other aviation groups, and individuals. Because of the increasing need for worldwide weather services, foreign weather organizations also provide vital input.

While weather forecasts are not 100 percent accurate, meteorologists, through careful scientific study and computer modeling, have the ability to predict weather patterns, trends, and characteristics with increasing accuracy. Through a complex system of weather services, government agencies, and independent weather observers, pilots and other aviation professionals receive the benefit of this vast knowledge base in the form of up-to-date weather reports and forecasts. These reports and forecasts enable pilots to make informed decisions regarding weather and flight safety before and during a flight.

Symbol	Meaning
R	Rain
RW	Rain Shower
S	Snow
SW	Snow Shower
TS	Thunderstorm
Intensity	
(none)	Light
+	Moderate
++	Heavy
X	Very Heavy
XX	Intense
	Extreme
Contraction	Operational Status
PPINE	Radar is operating normally but there are echoes being detected.
PPINA	Radar observation is not available.
PPIOM	Radar is inoperative or out of service.
AUTO	Automated radar report from WSR-88D.



Observations

The data gathered from surface and upper altitude observations form the basis of all weather forecasts, advisories, and briefings. There are four types of weather observations: surface, upper air, radar, and satellite.

Surface Aviation Weather Observations

Surface aviation weather observations (METARs) are a compilation of elements of the current weather at individual ground stations across the United States. The network is made up of government and privately contracted facilities that provide continuous up-to-date weather information. Automated weather sources, such as the Automated Weather Observing Systems (AWOS), Automated Surface Observing Systems (ASOS), as well as other automated facilities, also play a major role in the gathering of surface observations.

Surface observations provide local weather conditions and other relevant information for a specific airport. This information includes the type of report, station identifier, date and time, modifier (as required), wind, visibility, runway visual range (RVR), weather phenomena, sky condition, temperature/dew point, altimeter reading, and applicable remarks. The information gathered for the surface observation may be from a person, an automated station, or an automated station that is updated or enhanced by a weather observer. In any form, the surface observation provides valuable information about individual airports around the country. Although the reports cover only a small radius, the pilot can generate a good picture of the weather over a wide area when many reporting stations are viewed together.

Air Route Traffic Control Center (ARTCC)

The Air Route Traffic Control Center (ARTCC) facilities are responsible for maintaining separation between flights conducted under instrument flight rules (IFR) in the en route structure. Center radars (Air Route Surveillance Radar (ARSR)) acquire and track transponder returns using the same basic technology as terminal radars. Earlier center radars displayed weather as an area of slashes (light precipitation) and Hs (moderate rainfall). Because the controller could not detect higher levels of precipitation, pilots had to be wary of areas showing moderate rainfall. Newer radar displays show weather as three shades of blue. Controllers can select the level of weather to be displayed. Weather displays of higher levels of intensity make it difficult for controllers to see aircraft data blocks, so pilots should not expect air traffic control (ATC) to keep weather displayed continuously.

Upper Air Observations

Observations of upper air weather are more challenging than surface observations. There are several methods by which upper air weather phenomena can be observed:

radiosonde observations, pilot weather reports (PIREPs), Aircraft Meteorological Data Relay (AMDAR) and the Meteorological Data Collection and Reporting System (MDCRS). A radiosonde is a small cubic instrumentation package that is suspended below a six foot hydrogen- or helium-filled balloon. Once released, the balloon rises at a rate of approximately 1,000 feet per minute (fpm). As it ascends, the instrumentation gathers various pieces of data, such as air temperature, moisture, and pressure, as well as wind speed and direction. Once the information is gathered, it is relayed to ground stations via a 300 milliwatt radio transmitter.

The balloon flight can last as long as 2 hours or more and can ascend to altitudes as high as 115,000 feet and drift as far as 125 miles. The temperatures and pressures experienced during the flight can be as low as -130 °F and pressures as low as a few thousandths of what is experienced at sea level.

Since the pressure decreases as the balloon rises in the atmosphere, the balloon expands until it reaches the limits of its elasticity. This point is reached when the diameter has increased to over 20 feet. At this point, the balloon pops and the radiosonde falls back to Earth. The descent is slowed by means of a parachute. The parachute aids in protecting people and objects on the ground. Each year over 75,000 balloons are launched. Of that number, 20 percent are recovered and returned for reconditioning. Return instructions are printed on the side of each radiosonde.

Pilots also provide vital information regarding upper air weather observations and remain the only real-time source of information regarding turbulence, icing, and cloud heights. This information is gathered and filed by pilots in flight. Together, PIREPs and radiosonde observations provide information on upper air conditions important for flight planning. Many domestic and international airlines have equipped their aircraft with instrumentation that automatically transmits in flight weather observations through the DataLink system.

The Aircraft Meteorological Data Relay (AMDAR) is an international program utilizing commercial aircraft to provide automated weather observations. The AMDAR program provides approximately 220,000-230,000 aircraft observations per day on a worldwide basis utilizing aircraft onboard sensors and probes that measure wind, temperature, humidity/water vapor, turbulence and icing data. AMDAR vertical profiles and en route observations provide significant benefits to the aviation community by enhancing aircraft safety and operating efficiency through improved weather analysis and forecasting. The AMDAR program also contributes to improved short and medium term numerical weather forecasts for a wide range of services including

severe weather, defense, marine, public weather and environmental monitoring. The information is down linked either via Very High Frequency (VHF) communications through the Aircraft Communications Addressing and Reporting System (ACARS) or via satellite link through the Aircraft to Satellite Data Acquisition and Relay (ASDAR).

The Meteorological Data Collection and Reporting System (MDCRS) is an automated airborne weather observation program that is used in the U.S. This program collects and disseminates real-time upper-air weather observations from participating airlines. The weather elements are down linked via ACARS and are managed by Aeronautical Radio, Inc. (ARINC) who then forwards them in Binary Universal Form for the Representation of Meteorological Data (BUFR) format to the NWS and in raw data form to the Earth Science Research Laboratory (ESRL) and the participating airline. More than 1,500 aircraft report wind and temperature data with some of these same aircraft also providing turbulence and humidity/water vapor information. In conjunction with avionics manufacturers, each participating airline programs their equipment to provide certain levels of meteorological data. The monitoring and collection of climb, en route, and descent data is accomplished through the aircraft's Flight Data Acquisition and Monitoring System (FDAMS) and is then transmitted via ACARS. When aircraft are out of ACARS range, reports can be relayed through ASDAR. However, in most cases, the reports are buffered until the aircraft comes within ACARS range, at which point they are downloaded.

Radar Observations

There are four types of radars which provide information about precipitation and wind.

1. The WSR-88D NEXRAD radar, commonly called Doppler radar, provides in-depth observations that inform surrounding communities of impending weather. Doppler radar has two operational modes: clear air and precipitation. In clear air mode, the radar is in its most sensitive operational mode because a slow antenna rotation allows the radar to sample the atmosphere longer. Images are updated about every 10 minutes in this mode.

Precipitation targets provide stronger return signals; therefore, the radar is operated in the Precipitation mode when precipitation is present. A faster antenna rotation in this mode allows images to update at a faster rate, approximately every 4 to 6 minutes. Intensity values in both modes are measured in dBZ (decibels of Z) and are depicted in color on the radar image. [Figure 13-1] Intensities are correlated to intensity terminology (phraseology) for ATC purposes. [Figures 13-2 and 13-3]

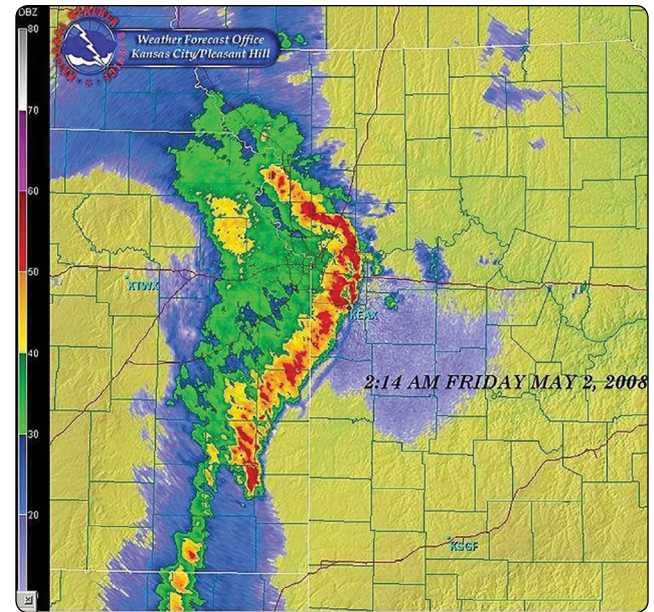


Figure 13-1. Example of a weather radar scope.

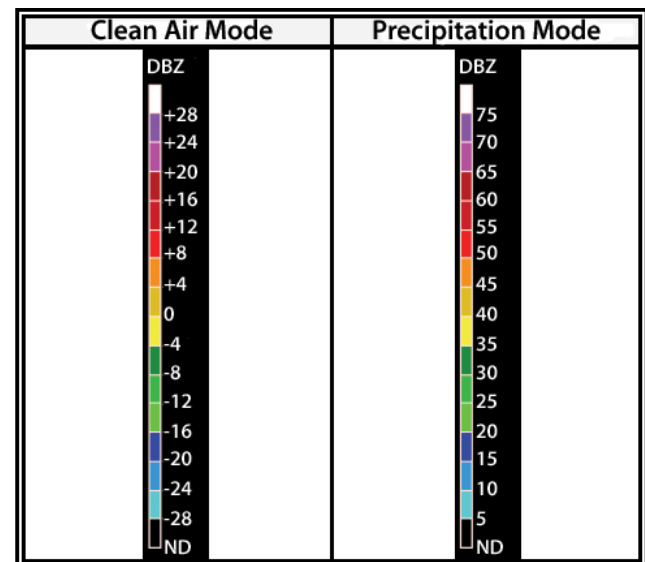


Figure 13-2. WSR-88D Weather Radar Echo Intensity Legend.

Reflectivity (dBZ) Ranges	Weather Radar Echo Intensity
<30 dBZ	Light
30–40 dBZ	Moderate
>40–50	Heavy
50+ dBZ	Extreme

Figure 13-3. WSR-88D Weather Radar Precipitation Intensity Terminology.

2. FAA terminal Doppler weather radar (TDWR), installed at some major airports around the country, also aids in providing severe weather alerts and warnings to ATC. Terminal radar ensures pilots are aware of wind shear, gust fronts, and heavy precipitation, all of which are dangerous to arriving and departing aircraft.
3. The third type of radar commonly used in the detection of precipitation is the FAA airport surveillance radar. This radar is used primarily to detect aircraft, but it also detects the location and intensity of precipitation, which is used to route aircraft traffic around severe weather in an airport environment.
4. Airborne radar is equipment carried by aircraft to locate weather disturbances. The airborne radars generally operate in the C or X bands (around 6 GHz or around 10 GHz, respectively) permitting both penetration of heavy precipitation, required for determining the extent of thunderstorms, and sufficient reflection from less intense precipitation.

Satellite

Advancement in satellite technologies has recently allowed for commercial use to include weather uplinks. Through the use of satellite subscription services, individuals are now able to receive satellite transmitted signals that provide near real-time weather information for the North American continent.

Service Outlets

Service outlets are government, government contract, or private facilities that provide aviation weather services. Several different government agencies, including the FAA, National Oceanic and Atmospheric Administration (NOAA), and the NWS work in conjunction with private aviation companies to provide different means of accessing weather information.

Flight Service Station (FSS)

The FSS is the primary source for preflight weather information. A preflight weather briefing from an FSS can be obtained 24 hours a day by calling 1-800-WX BRIEF from anywhere in the United States and Puerto Rico. Telephone numbers for FSS can be found in the Chart Supplement U.S. (formerly Airport/Facility Directory) or in the United States Government section of the telephone book.

The FSS also provides inflight weather briefing services and weather advisories to flights within the FSS area of responsibility.

Telephone Information Briefing Service (TIBS)

The Telephone Information Briefing Service (TIBS), provided by FSS, is a system of automated telephone

recordings of meteorological and aeronautical information. TIBS provides area and route briefings, airspace procedures, and special announcements. The recordings are automatically updated as changes occur. It is designed to be a preliminary briefing tool and is not intended to replace a standard briefing from a FSS specialist. The TIBS service can only be accessed by a touchtone phone. The phone numbers for the TIBS service are listed in the Chart Supplement U.S. (formerly Airport/Facility Directory).

Hazardous Inflight Weather Advisory Service (HIWAS)

Hazardous Inflight Weather Advisory Service (HIWAS), available in the 48 conterminous states, is an automated continuous broadcast of hazardous weather information over selected VOR navigational aids (NAVAIDs). The broadcasts include advisories such as AIRMETS, SIGMETS, convective SIGMETS, and urgent PIREPs. The broadcasts are automatically updated as changes occur. Pilots should contact a FSS or EFAS for additional information. VORs that have HIWAS capability are depicted on aeronautical charts with an “H” in the upper right corner of the identification box. [Figure 13-4]

Transcribed Weather Broadcast (TWEB) (Alaska Only)

A continuous automated broadcast of meteorological and aeronautical data over selected low or medium frequency (L/MF) and very high frequency (VHF) omnidirectional range (VOR) NAVAID facilities. The broadcasts are automatically updated as changes occur. The broadcast contains adverse conditions, surface weather observations, PIREPs, and a density altitude statement (if applicable). Recordings may also include a synopsis, winds aloft forecast, en route and terminal forecast data, and radar reports. At selected locations, telephone access to the TWEB has been provided (TEL-TWEB). Telephone numbers for this service are found

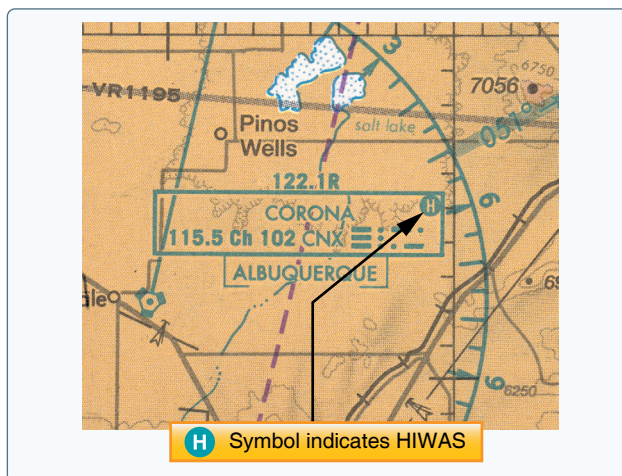


Figure 13-4. HIWAS availability is shown on sectional chart.

in the Alaska Chart Supplement U.S. (formerly Airport/Facility Directory). These broadcasts are made available primarily for preflight and inflight planning, and as such, should not be considered as a substitute for specialist-provided preflight briefings.

Weather Briefings

Prior to every flight, pilots should gather all information vital to the nature of the flight. This includes an appropriate weather briefing obtained from a specialist at a FSS.

For weather specialists to provide an appropriate weather briefing, they need to know which of the three types of briefings is needed—standard, abbreviated, or outlook. Other helpful information is whether the flight is visual flight rules (VFR) or IFR, aircraft identification and type, departure point, estimated time of departure (ETD), flight altitude, route of flight, destination, and estimated time en route (ETE).

This information is recorded in the flight plan system and a note is made regarding the type of weather briefing provided. If necessary, it can be referenced later to file or amend a flight plan. It is also used when an aircraft is overdue or is reported missing.

Standard Briefing

A standard briefing provides the most complete information and a more complete weather picture. This type of briefing should be obtained prior to the departure of any flight and should be used during flight planning. A standard briefing provides the following information in sequential order if it is applicable to the route of flight.

1. Adverse conditions—this includes information about adverse conditions that may influence a decision to cancel or alter the route of flight. Adverse conditions include significant weather, such as thunderstorms or aircraft icing, or other important items such as airport closings.
2. VFR flight not recommended—if the weather for the route of flight is below VFR minimums, or if it is doubtful the flight could be made under VFR conditions due to the forecast weather, the briefer may state “VFR flight not recommended.” It is the pilot’s decision whether or not to continue the flight under VFR, but this advisory should be weighed carefully.
3. Synopsis—an overview of the larger weather picture. Fronts and major weather systems that affect the general area are provided.
4. Current conditions—the current ceilings, visibility, winds, and temperatures. If the departure time is more than 2 hours away, current conditions are not included in the briefing.

5. En route forecast—a summary of the weather forecast for the proposed route of flight.
6. Destination forecast—a summary of the expected weather for the destination airport at the estimated time of arrival (ETA).
7. Forecast winds and temperatures aloft—a forecast of the winds at specific altitudes for the route of flight. The forecast temperature information aloft is provided only upon request.
8. Notices to Airmen (NOTAM)—information pertinent to the route of flight that has not been published in the NOTAM publication. Published NOTAM information is provided during the briefing only when requested.
9. ATC delays—an advisory of any known ATC delays that may affect the flight.
10. Other information—at the end of the standard briefing, the FSS specialist provides the radio frequencies needed to open a flight plan and to contact EFAS. Any additional information requested is also provided at this time.

Abbreviated Briefing

An abbreviated briefing is a shortened version of the standard briefing. It should be requested when a departure has been delayed or when weather information is needed to update the previous briefing. When this is the case, the weather specialist needs to know the time and source of the previous briefing so the necessary weather information is not omitted inadvertently. It is always a good idea for the pilot to update the weather information whenever he/she has additional time.

Outlook Briefing

An outlook briefing should be requested when a planned departure is 6 hours or more away. It provides initial forecast information that is limited in scope due to the time frame of the planned flight. This type of briefing is a good source of flight planning information that can influence decisions regarding route of flight, altitude, and ultimately the go/no-go decision. A prudent pilot requests a follow-up briefing prior to departure since an outlook briefing generally only contains information based on weather trends and existing weather in geographical areas at or near the departure airport. A standard briefing near the time of departure ensures that the pilot has the latest information available prior to his/her flight.

Aviation Weather Reports

Aviation weather reports are designed to give accurate depictions of current weather conditions. Each report provides current information that is updated at different times. Some typical reports are METARs and PIREPs.

Aviation Routine Weather Report (METAR)

A METAR is an observation of current surface weather reported in a standard international format. While the METAR code has been adopted worldwide, each country is allowed to make modifications to the code. Normally, these differences are minor but necessary to accommodate local procedures or particular units of measure. This discussion of METAR covers elements used in the United States.

METARs are issued on a regularly scheduled basis unless significant weather changes have occurred. A special METAR (SPECI) can be issued at any time between routine METAR reports.

Example:

METAR KGGG 161753Z AUTO 14021G26KT 3/4SM
+TSRA BR BKN008 OVC012CB 18/17 A2970 RMK
PRESFR

A typical METAR report contains the following information in sequential order:

1. Type of report—there are two types of METAR reports. The first is the routine METAR report that is transmitted on a regular time interval. The second is the aviation selected SPECI. This is a special report that can be given at any time to update the METAR for rapidly changing weather conditions, aircraft mishaps, or other critical information.
2. Station identifier—a four-letter code as established by the International Civil Aviation Organization (ICAO). In the 48 contiguous states, a unique three-letter identifier is preceded by the letter “K.” For example, Gregg County Airport in Longview, Texas, is identified by the letters “KGGG,” K being the country designation and GGG being the airport identifier. In other regions of the world, including Alaska and Hawaii, the first two letters of the four-letter ICAO identifier indicate the region, country, or state. Alaska identifiers always begin with the letters “PA” and Hawaii identifiers always begin with the letters “PH.” Station identifiers can be found by calling the FSS, a NWS office, or by searching various websites such as DUATS and NOAA's Aviation Weather Aviation Digital Data Services (ADDS).
3. Date and time of report—depicted in a six-digit group (161753Z). The first two digits are the date. The last four digits are the time of the METAR/SPECI, which is always given in coordinated universal time (UTC). A “Z” is appended to the end of the time to denote the time is given in Zulu time (UTC) as opposed to local time.
4. Modifier—denotes that the METAR/SPECI came from an automated source or that the report was corrected. If the notation “AUTO” is listed in the METAR/SPECI, the report came from an automated source. It also lists “AO1” (for no precipitation discriminator) or “AO2” (with precipitation discriminator) in the “Remarks” section to indicate the type of precipitation sensors employed at the automated station.

When the modifier “COR” is used, it identifies a corrected report sent out to replace an earlier report that contained an error (for example: METAR KGGG 161753Z COR).
5. Wind—reported with five digits (14021KT) unless the speed is greater than 99 knots, in which case the wind is reported with six digits. The first three digits indicate the direction the true wind is blowing from in tens of degrees. If the wind is variable, it is reported as “VRB.” The last two digits indicate the speed of the wind in knots unless the wind is greater than 99 knots, in which case it is indicated by three digits. If the winds are gusting, the letter “G” follows the wind speed (G26KT). After the letter “G,” the peak gust recorded is provided. If the wind direction varies more than 60° and the wind speed is greater than six knots, a separate group of numbers, separated by a “V,” will indicate the extremes of the wind directions.
6. Visibility—the prevailing visibility ($\frac{3}{4}$ SM) is reported in statute miles as denoted by the letters “SM.” It is reported in both miles and fractions of miles. At times, runway visual range (RVR) is reported following the prevailing visibility. RVR is the distance a pilot can see down the runway in a moving aircraft. When RVR is reported, it is shown with an R, then the runway number followed by a slant, then the visual range in feet. For example, when the RVR is reported as R17L/1400FT, it translates to a visual range of 1,400 feet on runway 17 left.
7. Weather—can be broken down into two different categories: qualifiers and weather phenomenon (+TSRA BR). First, the qualifiers of intensity, proximity, and the descriptor of the weather are given. The intensity may be light (–), moderate (), or heavy (+). Proximity only depicts weather phenomena that are in the airport vicinity. The notation “VC” indicates a specific weather phenomenon is in the vicinity of five to ten miles from the airport. Descriptors are used to describe certain types of precipitation and obscurations. Weather phenomena may be reported as being precipitation, obscurations, and other phenomena, such as squalls or funnel clouds.